CONNEXUS black sea tech forum | 2026

Center of competence “Digitalization of the economy in Big Data ecosystem”

Digitalization and AI approach

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Проект „Дигитализация на икономиката в среда на големи данни“
BG16RFPR002-1.014-0013-C01

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ICT CLUSTER VARNA

BSMEPA BULGARIAN SMALL AND MEDIUM ENTERPRISES PROMOTION AGENCY

ВАРНА

BRAIN++ AI FACTORY

ARC FUND

The Project “DIBD” has 6 partners

- University of National and World Economy – Sofia - Base Leading organization
- Plovdiv University “Paisii Hilendarski”
- Trakia university – St. Zagora
- Technical University – Gabrovo
- “Angel Kanchev” University of Ruse
- Institute of Information and Communication Technologies - BAS





Scientists from all over Bulgaria

Main goals of the Center of Competence

- Creation of **Scientific Infrastructure** for Digitization of Economy in Big Data environment

- Carrying out **Research Services** in the field of:
  - digitalization of the economy (digitalization of business processes)
  - in a Big Data environment

- Performing **Scientific Applied Services** in the field of digitization of the economy in a Big Data environment



What is Big data:

(not only big in size – GB, TB, PB)

- Structured data (for example Tables)
- **Semi-structured data (for example email, e-docuemnts)**
- **Fully unstructured data(for example graphics, pictures, video, audio)**

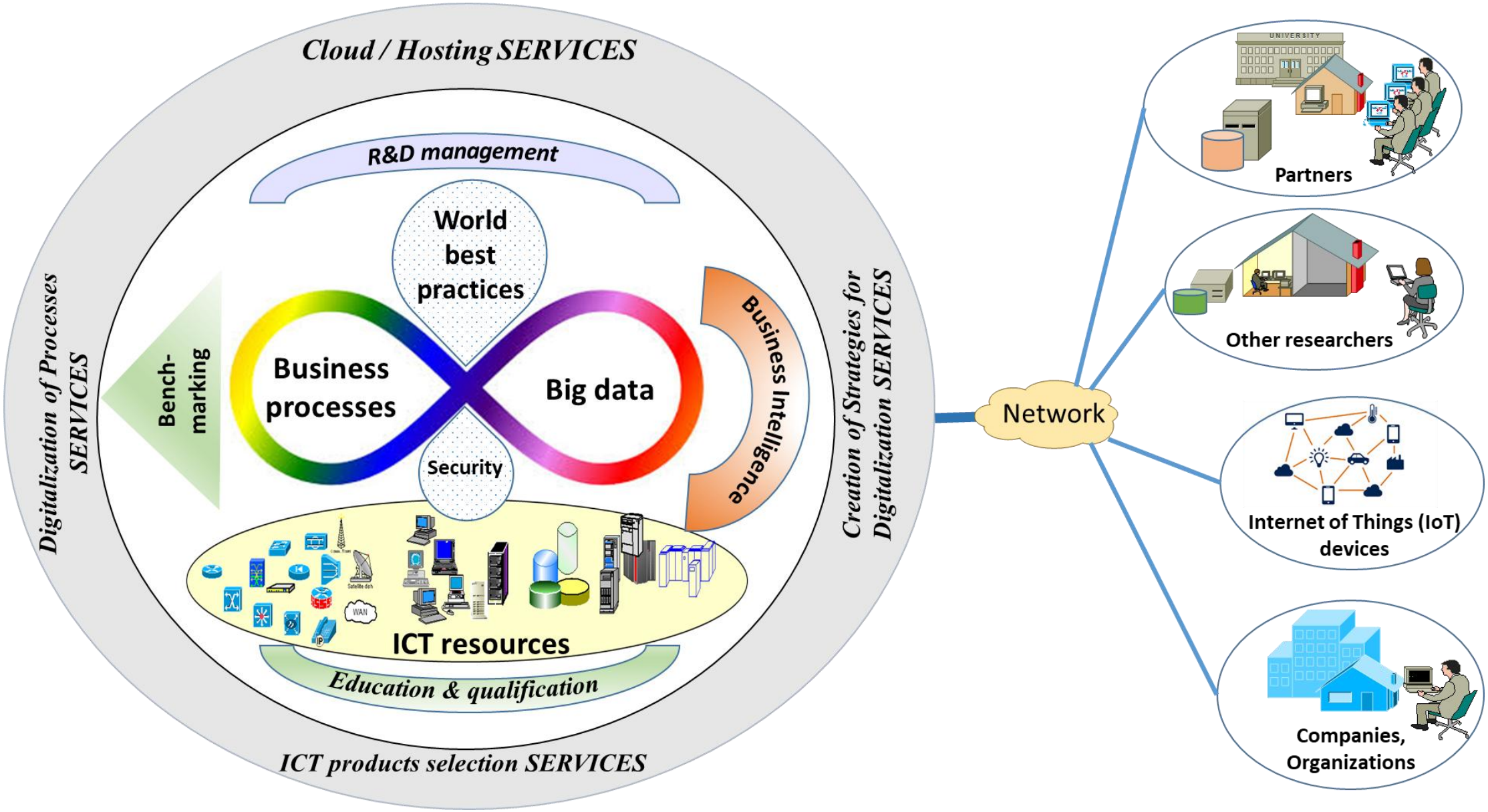
The Center of Competence covers 8 areas of the Economy

#	BUSINESS AREAS - Area of the Economy	Leading partner
1	Accounting	UNWE
2	Financial management	UNWE
3	Logistics and SCM	UNWE, Tr Uni
4	Marketing and CRM	UNWE
5	Human Resources management	UNWE
6	Real Estate management	EU Varna
7	Manufacturing	TU Gabrovo
8	IoT and Stock control	RU

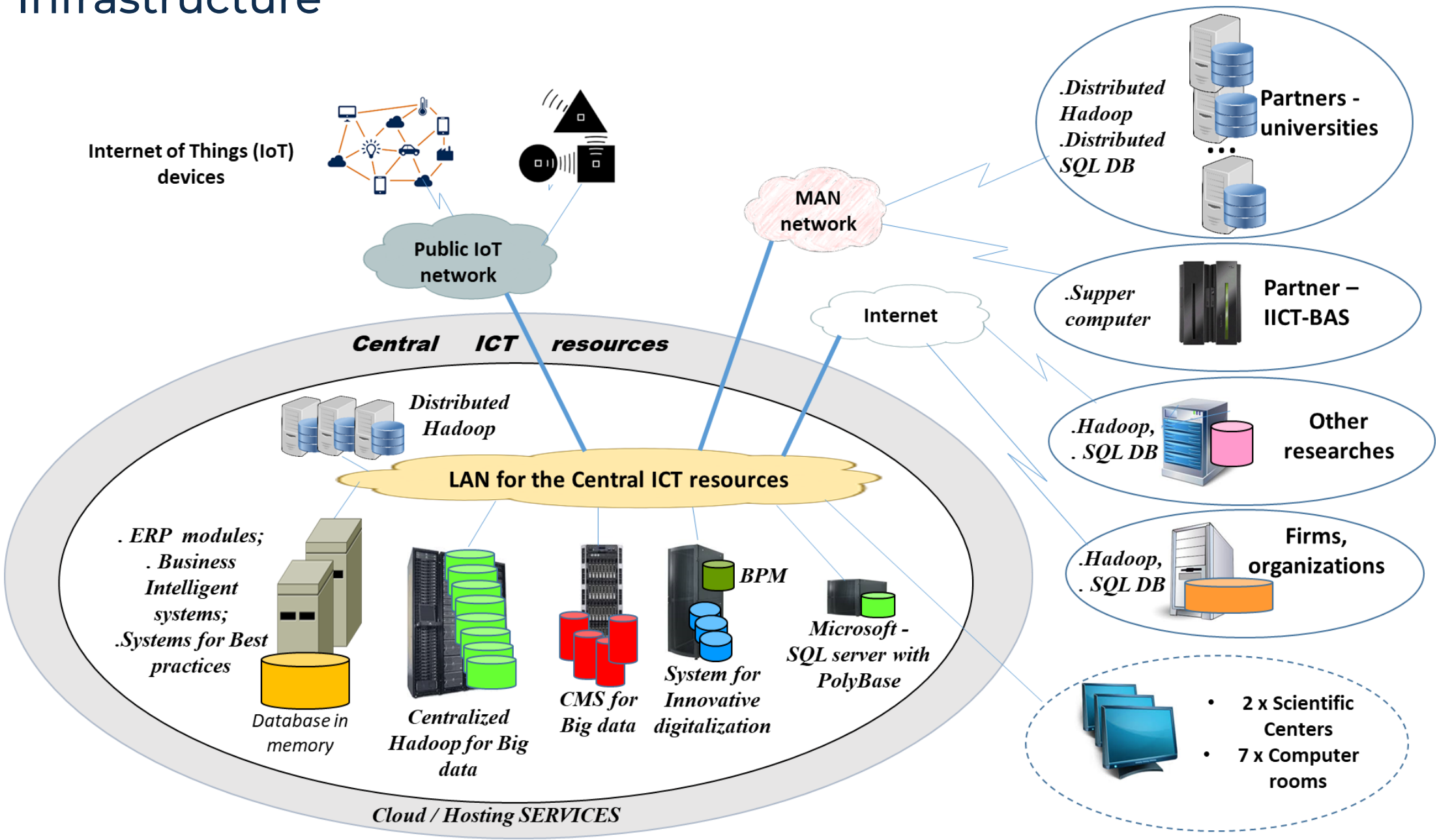
Additional area of Impact - Macroeconomics



Functional Architecture of the CoC's Infrastructure

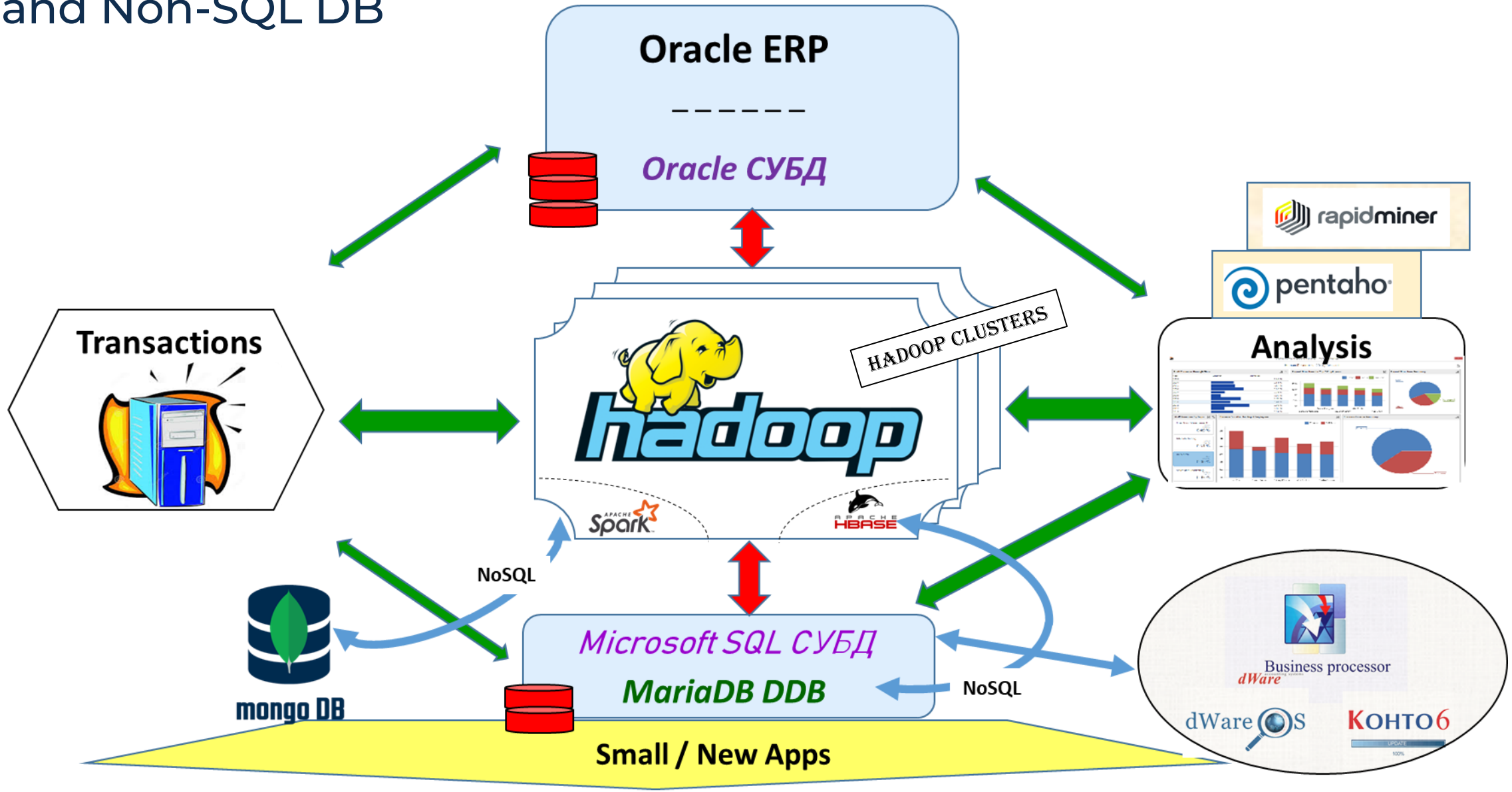


Conceptual ICT architecture of the CoC's Infrastructure



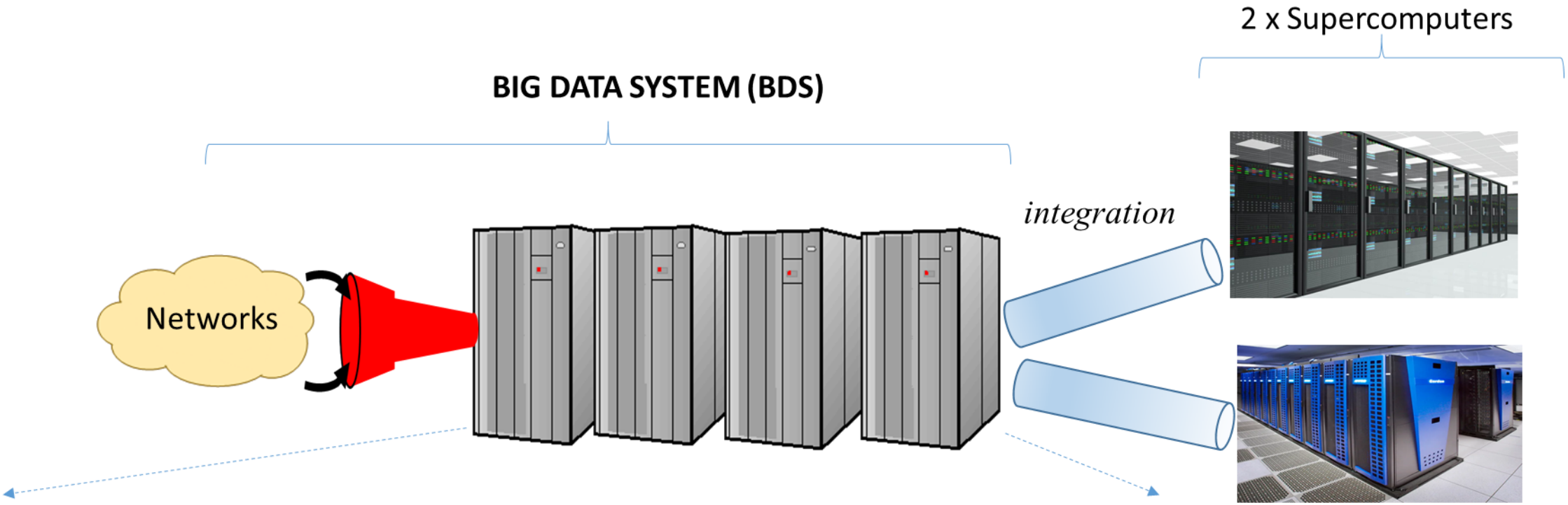
Digital business processes - Transactions and Analytics are served by Oracle DBMS, Microsoft DBMS, MariaDB DBMS and Non-SQL DB

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Big data system is integrated with the 2 Bulgarian Supercomputers

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Big Data System as **an Object** of Cyber Security

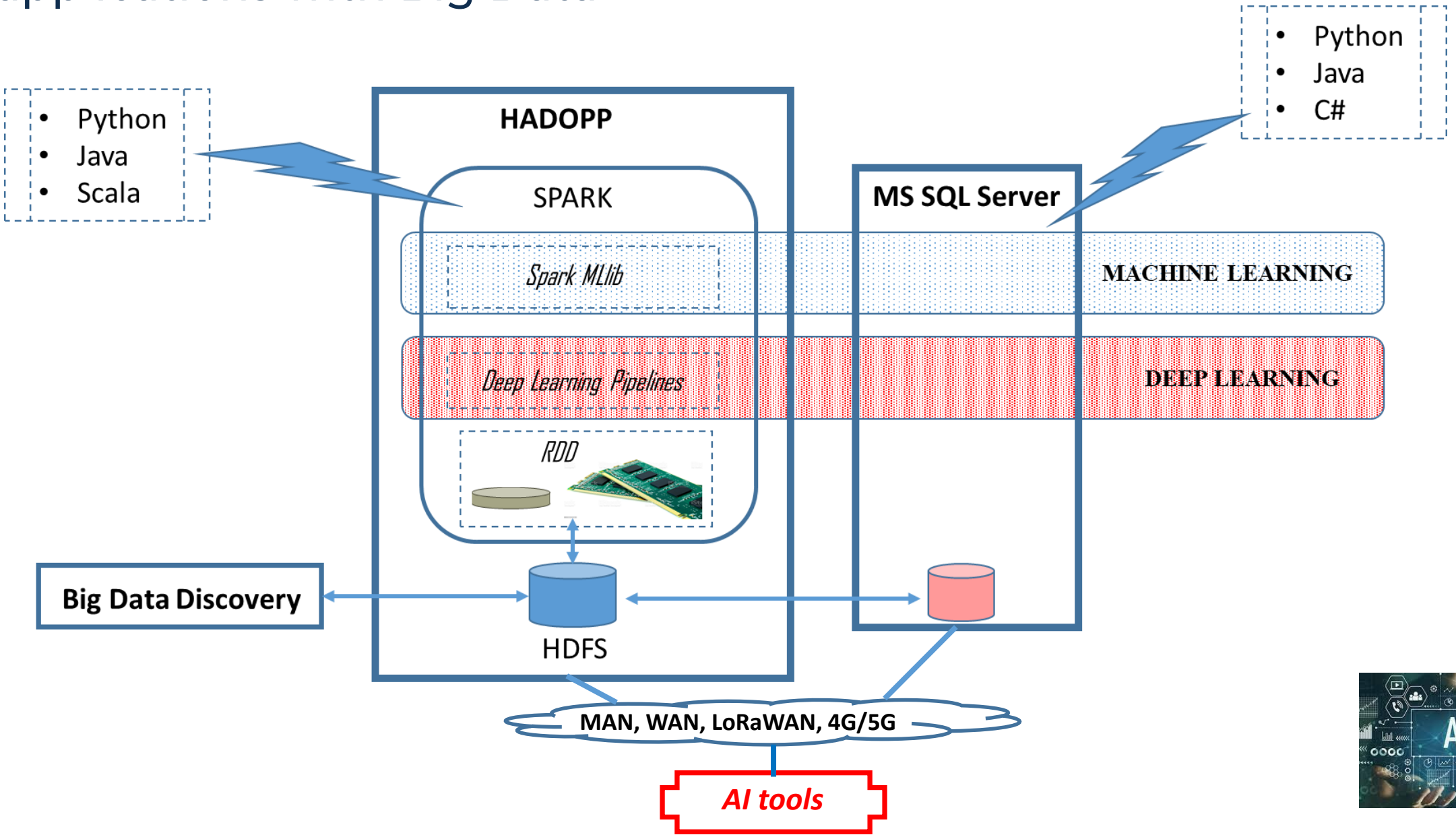
- Ensures security when data is received in the BDS
- Ensures security of data storage in BDS
- Ensures security of processing in the BDS

Big Data System as **an Instrument** of Cyber Security

- It ensures the security of receiving the data for Supercomputers
- Ensures security of data storage for Supercomputers
- It provides a large and reliable data repository for Supercomputers

Special focus is on building Artificial Intelligence applications with Big Data

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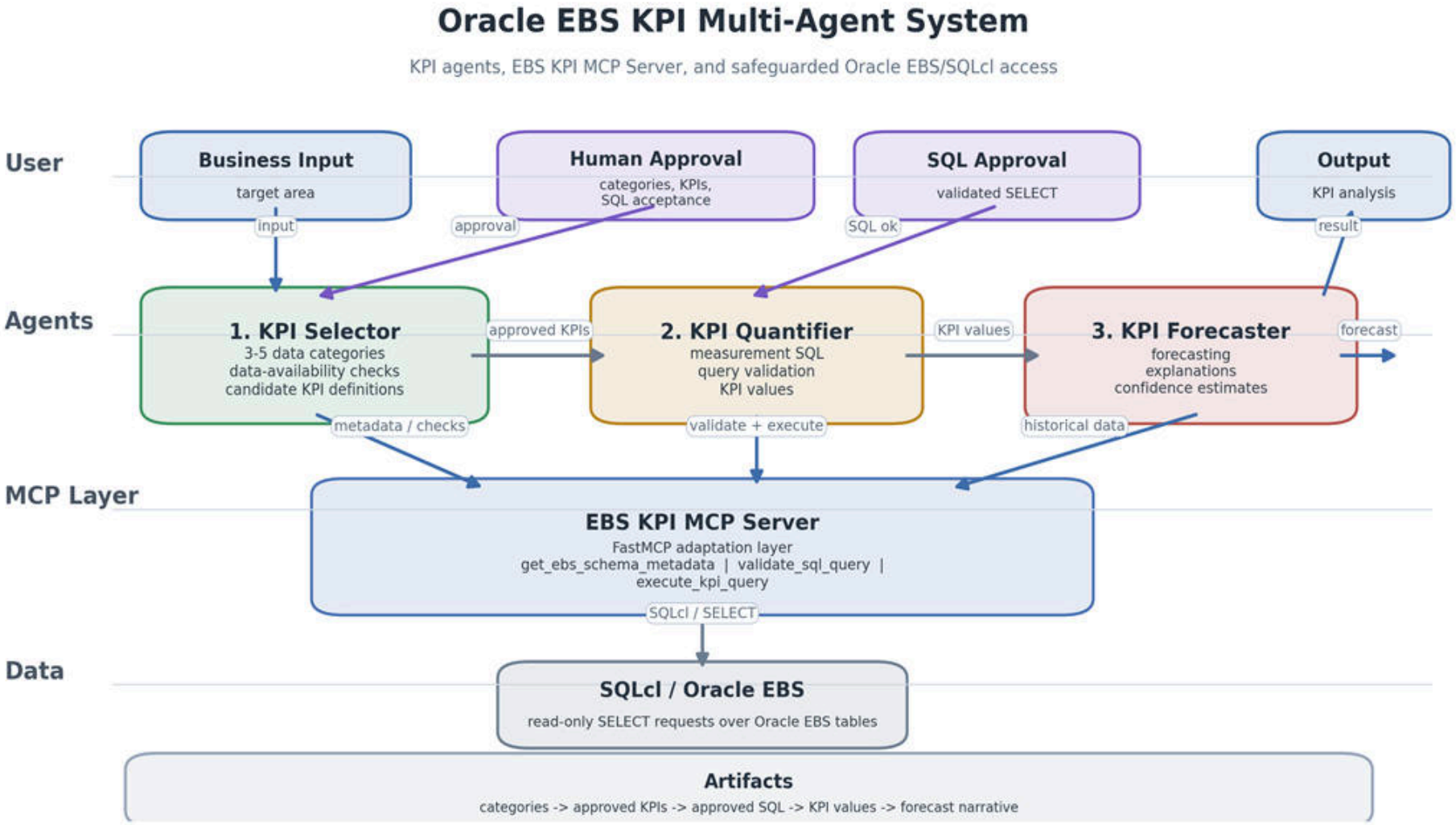


PoC#1 : Multi AI-agent system for dynamic KPIs over MCP based Oracle ERP (EBS)

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Coordinated LLM based workflow analytic layer

MCP for Oracle EBS

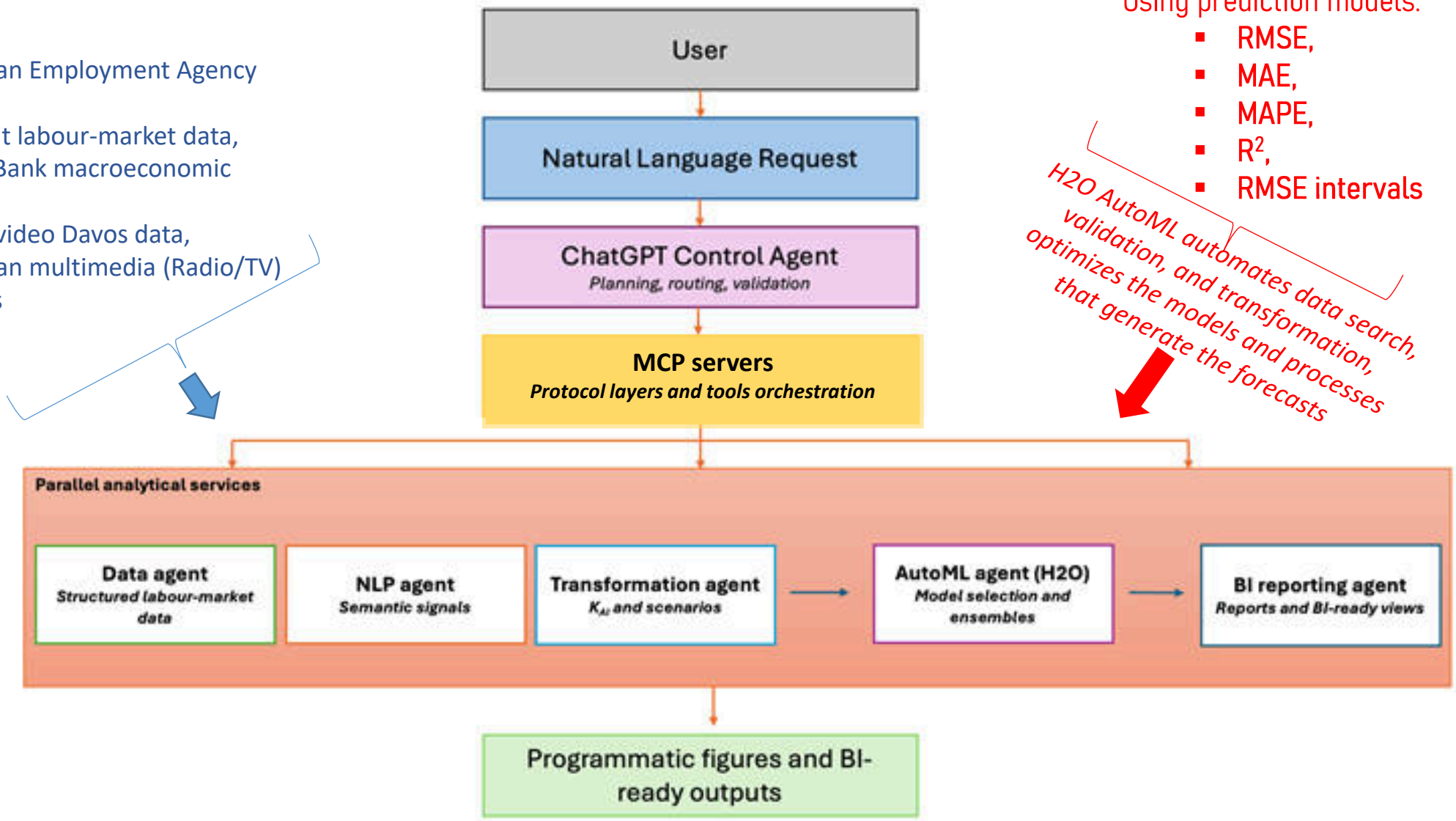


PoC#2: Multi AI-agent system for dynamic prediction of National and Regional unemployment Young (24-29 y.) Labor Markets in Bulgaria

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Uses data:

- Bulgarian Employment Agency data,
- Eurostat labour-market data,
- World Bank macroeconomic data,
- Audio/video Davos data,
- Bulgarian multimedia (Radio/TV) sources



Using prediction models:

- RMSE,
- MAE,
- MAPE,
- R^2,
- RMSE intervals

H2O AutoML automates data search, validation, and transformation, optimizes the models and processes that generate the forecasts

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Thank you !

Q & A

